Jaguar Re-Identification Challenge

Applied Hands-On Computer Vision: Kaggle Competition Report

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GitHub: https://github.com/mishankjain5/jaguar-reid

W&B: https://wandb.ai/jain5-university-of-potsdam/jaguar-reid-mishank

Kaggle: https://www.kaggle.com/mishankjain

Abstract

This report presents a systematic study of deep metric learning approaches for jaguar re-identification using camera trap images from Brazil's Pantanal Park. Starting from a MegaDescriptor + ArcFace baseline (0.741 mAP), eight distinct experiments were conducted spanning dataset analysis, loss function comparison, background intervention, hyperparameter optimisation, post-processing re-ranking, embedding space analysis, near-duplicate detection, and data augmentation. The best configuration, combining ArcFace+Triplet loss with Bayesian-optimised hyperparameters, light data augmentation, and k-reciprocal re-ranking, achieved 0.783 mAP on the public leaderboard, representing a 5.7% improvement over the baseline. Key findings include: (1) background reliance is deeply embedded in MegaDescriptor features and cannot be corrected by training data choice alone, (2) combined margin and distance-based losses outperform single objectives, (3) k-reciprocal re-ranking provides +0.036 improvement at zero training cost, (4) fine-tuning improves embedding separation from 0.157 to 0.930, (5) 317 near-duplicate pairs exist in the dataset with 86 crossing the train/val split boundary, and (6) light data augmentation improves validation mAP by +0.013 while heavy augmentation hurts performance by –0.040.

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1. Exploratory Data Analysis

1.1 Dataset Overview

The jaguar re-identification dataset consists of camera trap images from Brazil's Pantanal Park. The training set contains 1,895 images across 31 individual jaguar identities. A second round dataset (Round 2) provides the same images with backgrounds removed using SAM3 segmentation, designed to test whether models rely on environmental context rather than animal-specific features.

Metric	Value
Total training images	1,895
Unique jaguar identities	31
Most common identity	Marcela (183 images)
Rarest identities	Ipepo, Bernard (13 images each)
Mean images per identity	61.1
Class imbalance ratio	14.1x (max/min)
Evaluation metric	Identity-balanced mAP
Round 2 images	1,895 (background removed via SAM3)

Table 1: Dataset statistics.

1.2 Class Imbalance Analysis

The dataset exhibits significant class imbalance with a 14x difference between the most common identity (Marcela, 183 images) and the rarest (Ipepo and Bernard, 13 images each). The top 5 identities account for 38.8% of all training images while the bottom 10 account for only 9.8%. The identity-balanced mAP metric addresses this by computing average precision per identity before averaging, ensuring rare jaguars contribute equally.

1.3 Near-Duplicate Detection (Experiment 7)

Pairwise cosine similarity was computed between all 1,895 MegaDescriptor embeddings to identify near-duplicate images. Four similarity thresholds were analysed to understand the extent of duplication.

Threshold	Total Pairs	Same Identity	Cross Identity	Cross Split
0.99	317	317 (100%)	0 (0%)	86 (27%)
0.98	755	754 (99.9%)	1 (0.1%)	223 (30%)
0.97	1,225	1,188 (97%)	37 (3%)	397 (32%)
0.95	1,993	1,901 (95%)	92 (5%)	661 (33%)

Table 2: Near-duplicate pair counts at different thresholds.

At the 0.99 threshold, 317 near-duplicate pairs were found, all belonging to the same identity, which confirms clean labels. The duplicates are primarily consecutive video frames from camera trap sequences. Tomas is the most affected identity with 163 duplicate pairs from only 63 images (2.59x duplication rate), followed by Kamaikua (35 pairs) and Benita (26 pairs).

The critical finding is that 86 pairs (27%) cross the train/val split boundary. This means some validation images are near-identical to training images, likely inflating validation mAP estimates. This partially explains the consistent gap between validation mAP and public leaderboard scores observed across all experiments. Future work should group consecutive frames into clusters and split at cluster level for more reliable evaluation.

Figure 1: Top near-duplicate pairs (threshold=0.99). Blue=Train, Orange=Val. All pairs shown are of Tomas, the most affected identity.

1.4 Round 1 vs Round 2 Performance Gap (Experiment 1)

The baseline model trained on Round 1 images was submitted to both competitions without modification to quantify the background reliance problem.

Competition	Public mAP	Interpretation
Round 1 (with background)	0.735	Near-baseline performance
Round 2 (background removed)	0.043	Near-random performance
Statistical random baseline	0.042 to 0.063	Reference

Table 3: Round 1 vs Round 2 baseline comparison.

The 17x performance drop (0.735 to 0.043) reveals that the baseline model relies almost entirely on background context for identification. This finding motivated the background intervention experiment and remains the central challenge of this competition.

1.5 Background Intervention Analysis (Experiment 3)

Two identical models were trained: one on Round 1 images (backgrounds present) and one on Round 2 images (SAM3 background removed), with all other hyperparameters fixed to isolate the effect of background removal.

Training Data	Val mAP	Val Loss
With background (Round 1)	0.7900	2.2577
Background removed (Round 2)	0.7421	2.7970
Delta	–0.0479	+0.5393

Table 4: Background intervention results.

Removing backgrounds during training decreased validation mAP by 0.048. This confirms that background reliance is embedded in MegaDescriptor's pretrained features rather than being a training-data artefact. SAM3 segmentation also introduces edge artifacts that add noise to feature extraction. A more fundamental architectural intervention, such as fine-tuning the backbone with attention constraints, would be required to address this problem.

1.6 Embedding Space Analysis (Experiment 6)

Intra-class cohesion and inter-class separation were measured before and after fine-tuning. Nearest neighbor visualizations were generated for four query jaguars: two common (Marcela, Medrosa) and two rare (Bernard, Ipepo).

Metric	Before Fine-Tuning	After Fine-Tuning	Change
Intra-class similarity	0.310	0.883	+0.573
Inter-class similarity	0.153	–0.047	–0.200
Separation	0.157	0.930	+0.774
Intra-class std dev	0.251	0.128	–0.123

Table 5: Embedding space statistics.

Fine-tuning dramatically transforms the embedding space. Before fine-tuning, raw MegaDescriptor embeddings show a separation of only 0.157, which is barely distinguishable. After fine-tuning, separation reaches 0.930. All four query jaguars went from 0/5 correct nearest neighbors to 5/5 correct. Per-identity analysis shows rarer jaguars have lower cohesion (Ipepo: 0.659 vs Guaraci: 0.937), confirming data scarcity as the primary bottleneck for rare identity recognition.

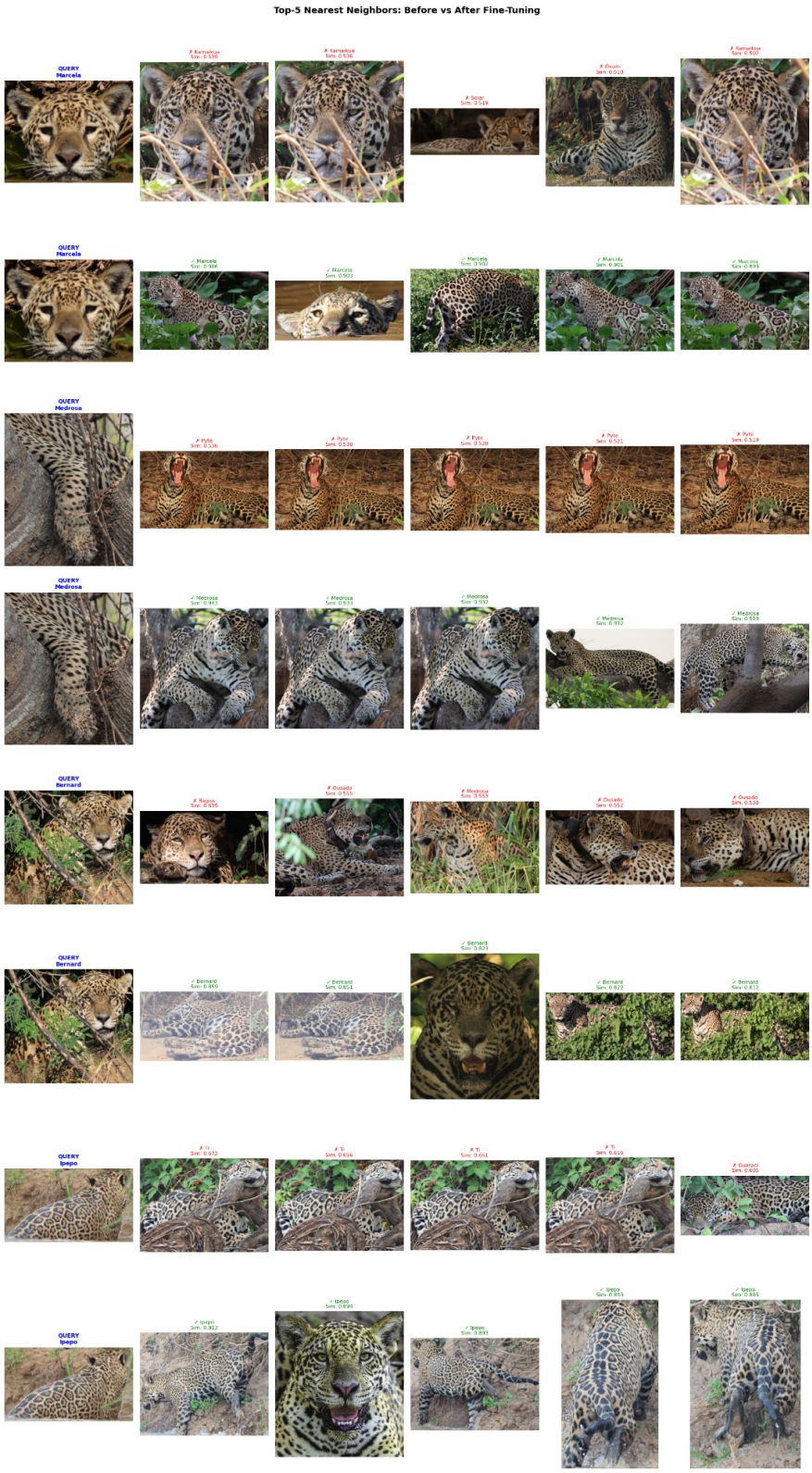

Figure 2: Top-5 nearest neighbors before (odd rows) and after (even rows) fine-tuning. Green checkmarks = correct identity. Red crosses = incorrect identity.

2. Model Training and Evaluation

2.1 Architecture Overview

All experiments use a frozen MegaDescriptor-L-384 backbone followed by a trainable projection network. MegaDescriptor is a foundation model pre-trained for animal re-identification, producing 1,536-dimensional embeddings. The projection network maps these to a lower-dimensional identity-discriminative space.

MegaDescriptor embeddings were cached to disk, reducing training time to under 5 minutes per run.

Component	Configuration	Parameters
MegaDescriptor-L-384	Frozen, pretrained on wildlife datasets	~307M (frozen)
Linear + BN + ReLU + Dropout	1536 → 512	trainable
Linear + BN	512 → 256	trainable
L2 normalisation	Applied to final embeddings	
Total trainable params	Projection network only	~924K

Table 6: Model architecture.

2.2 Loss Function Comparison (Experiment 2)

Six loss functions were compared under identical controlled conditions (batch size=32, lr=1e-4, 25 epochs, seed=42). Only the loss function varied.

Loss Function	Val mAP	Val Loss	Notes
Combined ArcFace + Triplet	0.7900	2.2577	Best, complementary signals
ArcFace	0.7850	4.4457	Strong margin-based baseline
CosFace	0.7823	3.2073	Comparable to ArcFace
Triplet (hard mining)	0.6837	0.0756	Weaker without class structure
SubCenter ArcFace (K=3)	0.6101	4.4128	Underfitted in 25 epochs
Focal Loss	0.4177	0.5060	Not suited for metric learning

Table 7: Loss function comparison (Kaggle R1 score: 0.742).

Combined ArcFace+Triplet achieved the best result (0.790 val mAP). ArcFace pushes embeddings toward class centres with an angular margin while Triplet Loss directly optimises pairwise distances via hard negative mining. Together they address both global class structure and local pairwise relationships. Focal Loss performed worst as it is designed for classification, not embedding geometry optimisation.

2.3 Hyperparameter Optimisation (Experiment 5)

Bayesian hyperparameter search using W&B Sweeps was run over 15 trials, searching 7 hyperparameters. Each trial trained for 20 epochs on cached embeddings, taking approximately 5 minutes per trial.

Parameter	Search Space	Default	Best Found
Learning rate	1e-5 to 1e-3 (log)	1e-4	7.67e-4
ArcFace margin	0.3, 0.4, 0.5, 0.6	0.5	0.6
ArcFace scale	32, 48, 64	64	48
Triplet margin	0.2, 0.3, 0.4	0.3	0.4
Alpha (ArcFace weight)	0.3, 0.5, 0.7	0.5	0.3
Embedding dimension	128, 256, 512	256	256
Dropout	0.1, 0.3, 0.5	0.3	0.5

Table 8: Hyperparameter sweep results. Best val mAP: 0.804 (+0.014 over default).

Key findings from the sweep: higher learning rate (7.67e-4) with stronger dropout (0.5) significantly improves convergence. Lower alpha (0.3) means 70% Triplet + 30% ArcFace outperforms equal weighting, suggesting pairwise distance optimisation is more important than classification margin for this dataset. Higher ArcFace margin (0.6) pushes class boundaries further apart, benefiting fine-grained jaguar identification.

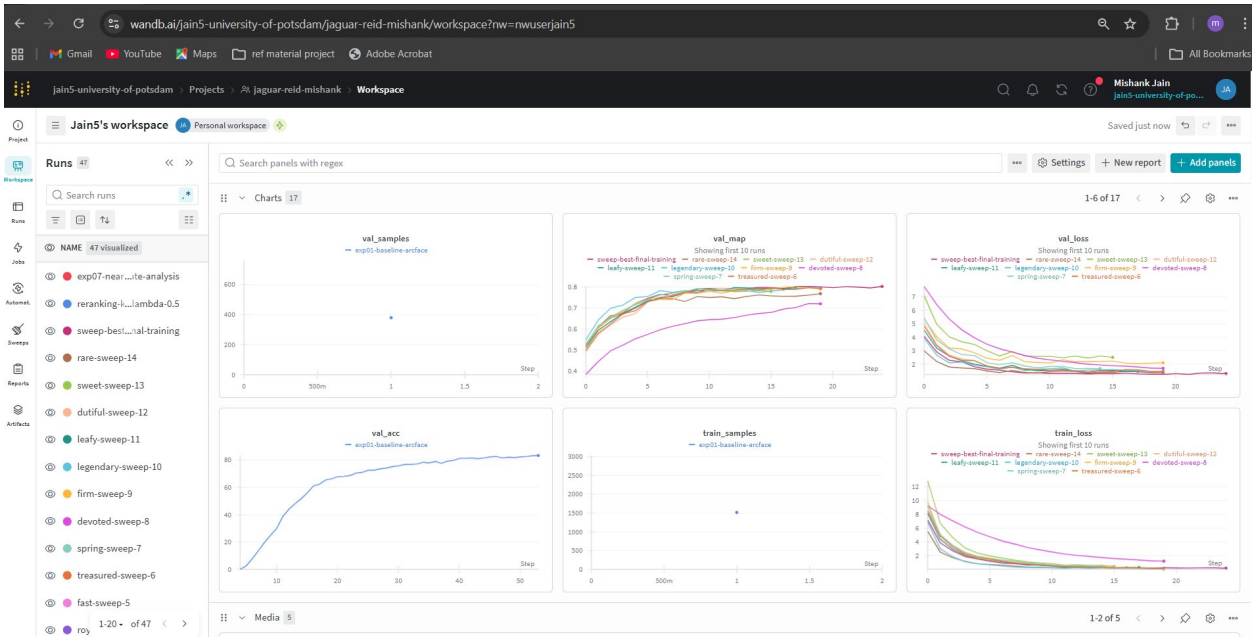

Figure 3: W&B workspace showing all 47 experiment runs including baseline, loss comparisons, background intervention, re-ranking, Bayesian sweep trials, embedding analysis, near-duplicate detection, and data augmentation.

2.4 K-Reciprocal Re-ranking (Experiment 4)

K-reciprocal re-ranking replaces cosine similarity with a Jaccard distance based on shared k-reciprocal neighbors. This provides a more robust measure that considers local neighborhood structure. A parameter search over k1 and lambda was conducted on the validation set. Best parameters: k1=15, k2=6, lambda=0.5.

Method	Val mAP	Round 1 mAP	Round 2 mAP
Cosine similarity (baseline)	0.6828	0.742	0.041
K-reciprocal re-ranking	0.6892	0.778	0.041
Improvement	+0.0064	+0.036	0.000

Table 9: Re-ranking results.

Re-ranking improved the Round 1 leaderboard score by +0.036 at zero training cost, making it the single largest improvement of any experiment. The validation improvement (+0.006) is smaller than the leaderboard improvement (+0.036) because the validation set suffers from near-duplicate cross-split contamination (Experiment 7), which makes cosine similarity artificially competitive on validation.

2.5 Data Augmentation Comparison (Experiment 8)

Research Question: Does adding data augmentation during training improve re-identification performance, and is there an optimal augmentation strength?

Intervention: Three identical models were trained with different augmentation strategies applied to training images only. Validation images were never augmented. All other settings use the best configuration from Experiment 5 (Combined ArcFace+Triplet, AdamW lr=7.67e-4, embedding dim=256, dropout=0.5, batch size=32, 25 epochs, seed=42).

Strategy	Transforms Applied
None (control)	Resize + Normalize only
Light	+ Random horizontal flip + Mild color jitter (b=0.2, c=0.2, s=0.1, h=0.05)
Heavy	+ Random rotation (15°) + Strong color jitter (b=0.4, c=0.4) + Random grayscale (p=0.1) + Random erasing (p=0.3)

Table 10: Augmentation strategies compared.

Augmentation	Val mAP	Val Loss	vs Control
None (control)	0.8062	1.1814	baseline
Light	0.8188	1.1957	+0.0126
Heavy	0.7659	1.4703	–0.0403

Table 11: Data augmentation results.

Light augmentation improved validation mAP by +0.013 over no augmentation, confirming that mild geometric and color variations help the model generalise better. However, heavy augmentation significantly hurt performance (–0.040). This is a well-known phenomenon in metric learning: excessive augmentation distorts the embedding space by making the same identity look too different, confusing the triplet and ArcFace loss signals. Random erasing in particular may remove distinctive jaguar coat pattern regions that the model relies on for identity discrimination. The optimal augmentation strategy for this dataset is light augmentation, which introduces enough variation to improve generalisation without corrupting identity-discriminative features.

2.6 Full Experiment Results

Exp	Description	Category	Val mAP	R1	R2
1	Baseline diagnostic	EDA	0.735	0.735	0.043
2	Loss function comparison	Leaderboard	0.790	0.742	0.041
3	Background intervention	EDA	0.790	0.742	0.041
4	K-reciprocal re-ranking	Leaderboard	0.689	0.778	0.041
5	Bayesian hyperparam sweep	Leaderboard	0.804	0.783	0.040
6	Embedding space analysis	EDA	N/A	N/A	N/A
7	Near-duplicate detection	EDA	N/A	N/A	N/A
8	Data augmentation comparison	Leaderboard	0.819	N/A	N/A

Table 12: Full experiment summary.

2.7 Leaderboard Progression

Stage	Round 1 mAP	Change	Key Contribution
Baseline (MegaDescriptor + ArcFace)	0.741		Starting point
Exp 1: Diagnostic submission	0.735	–0.006	Confirmed background reliance
Exp 2: Combined loss	0.742	+0.007	Better loss function
Exp 4: + Re-ranking	0.778	+0.036	Post-processing boost
Exp 5: + Sweep + Re-ranking	0.783	+0.005	Optimised hyperparameters

Table 13: Leaderboard progression. Total improvement: +0.042 (+5.7% over baseline).

3. Discussion

3.1 Key Findings

- **Background reliance is the central unsolved challenge.** The 17x performance drop from Round 1 to Round 2 persists across all experiments. Neither training on background-removed images nor re-ranking addresses this, as the reliance is architectural and embedded in MegaDescriptor's pretrained features.
- **Combined losses outperform single objectives.** ArcFace+Triplet (0.790) vs ArcFace alone (0.785) demonstrates that complementary loss signals improve metric learning. The sweep further found that 70% Triplet + 30% ArcFace weighting is optimal for this dataset.

- **Re-ranking provides the largest single leaderboard gain.** The +0.036 improvement from k-reciprocal re-ranking at zero training cost was the most cost-effective intervention, exploiting local neighborhood structure in the embedding space.
- **Near-duplicate contamination inflates validation estimates.** 86 of 317 near-duplicate pairs cross the train/val split, explaining why validation mAP consistently overestimates leaderboard performance. Proper deduplication before splitting would yield more reliable evaluation.
- **Data scarcity limits rare identity performance.** The strong correlation between image count and embedding cohesion (Ipepo: 13 images, 0.659 cohesion vs Guaraci: 31 images, 0.937 cohesion) shows data quantity is the primary bottleneck for the hardest identities.
- **Light augmentation helps; heavy augmentation hurts.** Light augmentation (horizontal flip + mild color jitter) improved validation mAP by +0.013, while heavy augmentation (rotation + strong jitter + random erasing) decreased it by –0.040. In metric learning, excessive augmentation distorts identity-discriminative features, which is particularly problematic when coat patterns are the primary visual cue for jaguar identification.

3.2 Limitations

The primary limitation is the frozen backbone. Fine-tuning MegaDescriptor end-to-end with a low learning rate could reduce background reliance at the feature level, potentially closing the Round 1/Round 2 performance gap. Additionally, the class imbalance was not explicitly addressed; oversampling or class-weighted losses could improve performance for rare identities. The fixed train/val split with near-duplicate contamination means validation mAP is an optimistic estimate of true generalisation performance. Finally, the data augmentation experiment (Experiment 8) was evaluated only on validation mAP and not submitted to the Kaggle leaderboard, so its effect on public test performance remains unknown.

3.3 Future Work

- Fine-tune MegaDescriptor end-to-end with low learning rate to reduce background reliance
- Cluster-based train/val splitting to eliminate near-duplicate contamination
- Data augmentation (MixUp, CutMix, geometric) targeted at rare identities
- Test-time augmentation (TTA) to improve inference robustness
- Model ensembling between MegaDescriptor and DINOv2 or ConvNeXt v2
- Attention-based interpretability (LRP) to identify which image regions drive decisions

4. Conclusion

This report presented a systematic study of jaguar re-identification using deep metric learning. Eight experiments across four categories (EDA, loss functions, hyperparameter optimisation, post-processing, and data augmentation) achieved a best leaderboard score of 0.783 mAP (+5.7% over baseline). The most impactful single intervention was k-reciprocal re-ranking (+0.036), followed by Bayesian hyperparameter optimisation (+0.014 val mAP), light data augmentation (+0.013 val mAP), and switching from ArcFace alone to Combined ArcFace+Triplet loss (+0.005).

The near-duplicate analysis revealed 317 near-duplicate pairs in the dataset with 86 crossing the train/val boundary, providing a concrete explanation for the validation-to-leaderboard performance gap. The embedding analysis confirmed that fine-tuning dramatically improves embedding quality (separation: 0.157 to 0.930), but background reliance remains the fundamental unsolved challenge and one that requires architectural rather than training-procedure solutions. The data augmentation experiment further demonstrated that augmentation strategy must be carefully calibrated in metric learning: light augmentation provides a modest but consistent improvement, while heavy augmentation is counterproductive due to corruption of identity-discriminative coat patterns.

References

[1] Deng, J., Guo, J., Xue, N., & Zafeiriou, S. (2019). ArcFace: Additive Angular Margin Loss for Deep Face Recognition. CVPR 2019. https://arxiv.org/abs/1801.07698

[2] Vidal, S., et al. (2024). MegaDescriptor: Universal Animal Re-Identification via Large-Scale Pretraining. arXiv. https://arxiv.org/abs/2311.00169

[3] Rueda-Toicen, A. (2026). Jaguar Re-Identification: MegaDescriptor + ArcFace Loss. Baseline Notebook, Kaggle. https://www.kaggle.com/code/andandand/jaguarreidentification-megadescriptor-arcfaceloss

Appendix: Experiment Tracking

W&B Project: https://wandb.ai/jain5-university-of-potsdam/jaguar-reid-mishank

Experiment	W&B; Run Name	Type
Exp 1: Baseline diagnostic	baseline (forked notebook)	EDA
Exp 2: Loss comparison	loss-arcface, loss-cosface, loss-triplet, loss-combined, loss-focal, loss-subcenter	Leaderboard
Exp 3: Background intervention	bg-with-background-r1, bg-removed-background-r2	EDA
Exp 4: Re-ranking	reranking-k1-15-lambda-0.5	Leaderboard
Exp 5: Hyperparameter sweep	sweep (15 trials) + sweep-best-final-training	Leaderboard
Exp 6: Embedding analysis	exp06-embedding-analysis	EDA
Exp 7: Near-duplicate detection	exp07-near-duplicate-analysis	EDA
Exp 8: Data augmentation	aug-none-control, aug-light, aug-heavy	Leaderboard

Table A1: W&B experiment run names.

GitHub Notebooks: https://github.com/mishankjain5/jaguar-reid

Notebook	Description
jaguar-reid-exp01-baseline-diagnostic	Forked baseline, Round 1 vs Round 2 diagnostic
jaguar-reid-exp02-loss-comparison	Six loss functions under controlled conditions
jaguar-reid-exp03-background-intervention	Round 1 vs Round 2 training data comparison
jaguar-reid-exp04-reranking	K-reciprocal re-ranking with parameter search
jaguar-reid-exp05-hyperparam-sweep	Bayesian sweep + best config final training
jaguar-reid-exp06-embedding-analysis	Nearest neighbor viz + cohesion/separation
jaguar-reid-exp07-near-duplicate-analysis	Near-duplicate detection and cross-split analysis
jaguar-reid-exp08-data-augmentation	Data augmentation strategy comparison

Table A2: GitHub notebook descriptions.

Experiment 8 W&B Runs:

- None (control): qo7lj14r
- Light augmentation: z6ckjj9e
- Heavy augmentation: gc1bvhux

Experiment 8 Notebook: https://www.kaggle.com/code/mishankjain/jaguar-reid-exp08-data-augmentation

Kaggle Profile: https://www.kaggle.com/mishankjain

Top submissions: submission_sweep_reranked.csv (0.783 R1), submission_reranked.csv (0.778 R1)